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The mediating role of organizational justice on the relationship between administrative performance appraisal practices and organizational commitment

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This study investigates the relationships among administrative performance appraisal (PA) practices, perception of organizational justice and organizational commitment. The results obtained from 395 employees who work in manufacturing companies in Taiwan show that the implementation of administrative PA activities is highly associated with employee perception of organizational justice and that the level of perceived organizational justice is highly associated with the level of organizational commitment. The results also demonstrate that perceived organizational justice has a partial mediating effect on the relationship between administrative PA practices and organizational commitment. Theoretical and practical implications of these results are discussed.

Keywords: human resource practices; organizational commitment; organizational justice; performance appraisal activities; Taiwan

Introduction

Performance appraisal (PA) activities are among the most important human resource (HR) management practices in organizations insofar as they yield critical data that are then used to make key decisions with regard to various HR actions and outcomes (Murphy and Cleveland 1995). Organizations have long been carrying out such activities while at the same time researchers have been studying and trying to improve them (Levy and Williams 2004). One major aim of PA is to help managers make administrative decisions including salary adjustments and promotions, as well as employee retentions or terminations.

Since PA involves measuring the job performance of individuals (Murphy and Cleveland 1995), whether the measurement is perceived as fair or not is a critical one with regard to successful implementations. Previous studies (Jawahar 2007; Kavanagh, Benson and Brown 2007) indicated that employee participation in PA, satisfaction with PA and knowledge about the related processes are highly associated with employee perceptions of fairness. For example, Jawahar (2007), in his seminal work, found that distributive justice influences satisfaction with performance ratings and procedural justice influences satisfaction with the appraisal system.

In addition, some empirical evidence shows that perceived fairness, satisfaction with PA practices and the quality of PA practices can all be used to predict the level of organizational commitment among employees (Ogilvie 1986; Kinicki, Carson and Bohlander 1992; Meyer and Smith 2000; Kuvvas 2006, 2011; Brown, Hyatt and Benson 2010). According to social exchange theory, when a person does a favor to another, the

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recipient of the favor will be obliged to reciprocate (Gouldner 1960). Based on this reasoning, individuals will reciprocate sources of fairness through behaviors that benefit the source. Organizational outcomes, such as organizational commitment or organizational citizenship behavior (OCB), may be mechanisms through which individuals reciprocate organizational fairness (Erdogan 2002). For example, Folger and Konovsky (1989) showed that procedural justice explains a larger variance in organizational commitment compared to distributive justice. Kuvvas (2006) found that satisfaction with PA has a positive association with affective organizational commitment. Brown et al. (2010) revealed that employees with low quality PA experiences were more likely to be less committed to their organizations. A recent study conducted by Zhang and Agarwal (2009) indicated that the three types of organizational justice, distributive, procedural and interactional, were positively associated to OCB. It is also showed that perceptions of organizational justice had a mediating effect on the relationship between HR practices and OCB.

Although there have been many studies in the fields of administrative PA, organizational justice and organizational commitment as noted above, most examine the relationship between either only two of these, such as that between PA and organizational justice (e.g. Folger and Konovsky 1989), between organizational justice and organizational commitment (e.g. Konovsky and Cropanzano 1991), or between PA and organizational commitment (e.g. Kuvvas 2006, 2011). Furthermore, while researchers into organizational justice have differentiated between three distinct dimensions of justice (distributive, procedural and interactional), much of the related empirical work has examined only one or two types of organizational justice (Cohen-Charash and Spector 2001). For these reasons, there is a need to investigate all three constructs in a holistic way, thus providing a more comprehensive understanding of the relationships among them.

Therefore, the aim of this study is to test the relationships among employee recognition of PA activities, perceptions of organizational justice and the level of organizational commitment. In so doing, I first examine whether employee recognition of administrative PA practices has a direct effect on perceptions of organizational justice. Next, I test whether employee perceptions of organizational justice have a direct effect on organizational commitment. Finally, I analyze the mediating effect that perceived organizational justice has on the relationship between employee recognition of administrative PA practices and organizational commitment.

Theoretical development

Administrative PA practices

Based on the administrative purpose of PA, three key HRM practices were identified in this study: salary adjustments, promotion decisions and performance standards.

Salary adjustments

The salary an employee receives is associated with the contribution they make to the organization, and PA is the only way that managers can use to assess work dynamics that exist among staff. Several studies (Summers and Hendrix 1991; Anthony, Perrewe and Kacmar 1993; Milkovich and Newman 1993; Bloom 1999; Boswell and Boudreau 2000; Milliman, Nason, Zhu and De Cieri 2002) suggest that when the employees perceive the salary adjustments as being unfair, then this will lead to negative feeling about their organizations, and this is likely to decrease their organizational commitment.

Promotion decisions

The results of PA have been often used to determine whether an employee has the potential to be promoted or not (Milliman et al. 2002). Since only some employees can be promoted, this means that such decisions produce zero-sum results that can influence the employee perceptions of distributive justice (Wooten and Cobb 1999; Nurse 2005). Many researchers have suggested that organizational justice plays a role in the relationship between promotion decisions and employee reactions to them, given that employees are likely to have positive perceptions of their organization and organizational justice when they receive rewards, such as promotions, which in turn translates into higher levels of organizational attachment, for example organizational commitment and citizenship behavior, or lower level of intention to quit (Halaby and Sobel 1979; Beehr, Taber and Walsh 1980; Arvey and Sackett 1993; De Souza 2002; Bagdadli, Roberson and Paoletti 2006).

Performance standards

Establishing consistent performance standards is an essential aspect of effective PA, as unclear standards have a negative impact on perceived fairness (Landy, Barnes and Murphy 1978; Dipboye and de Pontbriand 1981; Greenberg 1986; DeNisi and Pritchard 2006). However, the performance standards in many organizations remain ill-defined (Schneier, Shaw and Beatty 1991). In addition, even when performance standards are defined clearly, employees will still perceive them as unfair if implemented inconsistently (Gosselin, Werner and Halle 1997; DeNisi and Pritchard 2006; Kavanagh et al. 2007).

Perceived PA fairness and organizational commitment

Recent research into PA has shifted its focus to employee perceptions of the fairness of such appraisals (Greenberg 1986; Folger, Konovsky and Cropanzano 1992; Murphy and Cleveland 1995; Taylor, Tracy, Renard, Harrison and Carroll 1995; Cawley, Keeping and Levy 1998; Pettijohn, Pettijohn and d'Amico 2001; Levy and Williams 2004; Brown and Heywood 2005; Nurse 2005; Cropanzano, Bowen and Gilliland 2007; Jawahar 2007; Kavanagh et al. 2007). For instance, Jawahar (2007) suggested that the employee perceptions of fairness of PA have a significant influence on the success and further development of an organization's appraisal system. Moreover, Cropanzano et al. (2007) indicated that the employee perceptions of PA fairness could influence their levels of organizational commitment, OCB or intention to quit.

According to equity theory (Adams 1963), organizational justice consists of three constructs, namely, distributive justice, procedural justice and interactional justice. Cropanzano et al. (2007) stated that distributive justice is related to how resources are allocated in organizations, with the key point being not that all employees are treated equally, but that each employee feels that the results of this distribution are fair. In contrast, procedural justice is related to the notion that the distribution process should be fair and appropriate, and it is not related to the actual results of this process. Finally, interactional justice is based on the interpersonal interactions that occur among employees, with an emphasis on whether people are treated with dignity, courtesy and respect (Bies and Moag 1986).

Research has shown that employee perceived organizational justice has a significant influence on organizational commitment (Alexander and Ruderman 1987; Folger and Konovsky 1989; Konovsky and Cropanzano 1991; McFarlin and Sweeney 1992; Sweeney

and McFarlin 1993; Lowe and Vondanovich 1995; Randall and Mueller 1995; Martin and Bennett 1996; Konovsky 2000; Masterson, Lewis, Goldman and Taylor 2000; Cardy and Dobbins 1986; Aryee, Budhwar and Chen 2002; Kernan and Hanges 2002; Simons and Roberson 2003; Hui and Rupp 2005; Olkkonen and Lipponen 2006; Lavelle, Rupp and Brockner 2007). Organizational commitment is a job attitude that Robbins and Judge (2009) defined as ‘a state in which an employee identifies with a particular organization and its goals and wishes to maintain membership in the organization’ (pp. 113). Meyer and Allen (1991), and Meyer, Allen and Smith (1993) stated that the three constructs of affective, continuance and normative commitment together make up the overall concept of organizational commitment, and these have been widely applied in the literature. Affective commitment is the emotional attachment to the organization that an employee feels, as well as a belief in its values. Continuance commitment is the perceived economic value of remaining with an organization compared to leaving it. Finally, normative commitment is an obligation to remain with the organization for moral or ethical reasons.

Although many studies regarding the relationship between organizational justice and organizational commitment have been conducted in past few decades, most focused on the influence of specific PA practices on employee perceptions of organizational justice and their related levels of organizational commitment. In order to address some of the gaps in the current literature, this work develops a research model (Figure 1) to explore the relationships among PA practices, organizational justice and organizational commitment, as explained in more detail in the following sections.

Salary adjustments and organizational justice

Folger and Konovsky (1989) proposed that salary adjustment decisions that are made based on an employee’s actual performance can positively affect perceptions of distributive justice. They also indicated that if such decisions are carried out following systematic, standardized and precise processes, then this is also likely to improve perceived procedural justice. Other studies (Giles and Mossholder 1990; Milkovich and

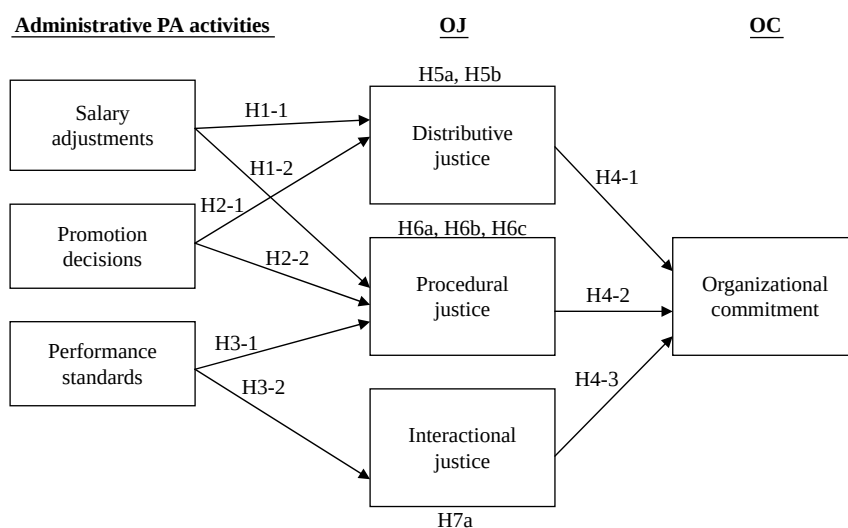


Figure 1. Research model.

Newman 1993; Boswell and Boudreau 2000; Tekleab, Bartol and Liu 2005) have also found that salary adjustment decisions are strongly associated with perceived organizational justice. Therefore, this work examines the relationships between the PA activity of salary adjustment and perceived organizational justice using the following hypotheses:

Hypothesis 1.1: Salary adjustments have a positive influence on distributive justice.

Hypothesis 1.2: Salary adjustments have a positive influence on procedural justice.

Promotion decisions and organizational justice

If unfair and subjective evaluation systems are used to determine the outcomes of promotion decisions, then this is likely to negatively affect employee perceptions of distributive and procedural justice (Mount 1983; Greenberg 1986; Colquitt et al. 2001; Nurse 2005; Tekleab et al. 2005; Jawahar 2007). For example, Greenberg (1986) used an open-question survey of managers in order to clarify the key events used to decide salary increases and promotion opportunities, and found that such decisions should be based on employees' actual performance, so that the rewards in an organization can be fairly distributed. He also found that employees should have channels of communication to challenge or appeal such decisions. Therefore, I examine the relationship between the PA activity of promotion decisions and perception of organizational justice, as follows:

Hypothesis 2.1: Promotion decisions have a positive influence on distributive justice.

Hypothesis 2.2: Promotion decisions have a positive influence on procedural justice.

Performance standards and organizational justice

Erdogan (2002) proposed that the more objective standards that PA follows, the greater the fairness perceived by employees. Other studies (Folger et al. 1992; Tang and Sarsfield-Baldwin 1996; Williams and Levy 2000) also show that adequate notice of performance standards is important to employee perceptions of procedural fairness, and this means giving employees knowledge of the appraisal system used and how it affects them well ahead of any formal PA (Narcisse and Harcourt 2008). These standards must be well documented, clearly explained, fully understood and preferably set by mutual agreement, with employees only held accountable for those that have been properly communicated to them. Prior research also indicated that interpersonal factors between managers and subordinates can influence the results of PA (Alexander and Wilkins 1982; Bowen, Gilliland and Folger 1999; Cardy and Dobbins 1986; Tsui and Barry 1986; Varma, DeNisi and Peters 1996). For instance, Tsui and Barry (1986) found that appraisers who have positive feelings about their employees tend to give positive performance evaluations to them. While no studies have yet examined interactional justice in the context of performance standards, Cropanzano, Prehar, and Chen (2002) found that it is associated with the perceived quality of treatment received from supervisors. This implies that if employees receive more positive treatment from supervisors, they will have more opportunities to communicate with them and thus receive more adequate notice of performance standards. I thus argue that employees' knowledge of performance standards is correlated to their perceptions of procedural and interactional justice, and so the following hypotheses are proposed:

Hypothesis 3.1: An employee's knowledge of performance standards has a positive influence on procedural justice.

Hypothesis 3.2: An employee's knowledge of performance standards has a positive influence on interactional justice.

Organizational justice and organizational commitment

Numerous studies have examined the relationship between organizational justice and organizational commitment and found that distributive justice is highly associated with continuance or normative commitment (Alexander and Ruderman 1987; Lowe and Vondanovich 1995; Randall and Mueller 1995; Fields, Pang and Chiu 2000; Konovsky 2000). Prior research also found that procedural justice is positively related to continuance, normative or affect commitment (Konovsky and Cropanzano 1991; Lowe and Vondanovich 1995; Fields et al. 2000; Masterson et al. 2000; Kernan and Hanges 2002; Kickul, Lester and Finkl 2002; Simons and Roberson 2003; Hui and Rupp 2005). Furthermore, previous research indicated that interactional justice has positive influences on continuance or affect commitment (Simons and Roberson 2003). Therefore, I propose that employees' perceived organizational justice would influence their organizational commitment, and so hypothesize that:

Hypothesis 4: An employee's perceived organizational justice has a positive influence on organizational commitment.

Hypothesis 4.1: An employee's perceived distributive justice has a positive influence on organizational commitment.

Hypothesis 4.2: An employee's perceived procedure justice has a positive influence on organizational commitment.

Hypothesis 4.3: An employee's perceived interactional justice has a positive influence on organizational commitment.

Mediation

The relationships outlined above might be better represented through a mediation model. According to Zhang and Agarwal (2009), organizational justice acts as a mediator between HR practices and workplace outcomes. Based on the findings of earlier studies and a combination of the theories of organizational justice and organizational commitment described previously, this work anticipates that the three administrative PA practices presented above can be used to predict perceived organizational justice, which in turn can predict organizational commitment. I thus hypothesize that:

Hypothesis 5: Employee perceptions of distributive justice mediate the relationship between administrative PA practices and organizational commitment.

Hypothesis 5.1: Employee perceptions of distributive justice mediate the relationship between salary adjustments and organizational commitment.

Hypothesis 5.2: Employee perceptions of distributive justice mediate the relationship between promotion decisions and organizational commitment.

Hypothesis 6: Employee perceptions of procedural justice mediate the relationship between administrative PA practices and organizational commitment.

- Hypothesis 6.1:* Employee perceptions of procedural justice mediate the relationship between salary adjustments and organizational commitment.
- Hypothesis 6.2:* Employee perceptions of procedural justice mediate the relationship between promotion decisions and organizational commitment.
- Hypothesis 6.3:* Employee perceptions of procedural justice mediate the relationship between performance standards and organizational commitment.
- Hypothesis 7:* Employee perceptions of interactional justice mediate the relationship between administrative PA practices and organizational commitment.
- Hypothesis 7.1:* Employee perceptions of interactional justice mediate the relationship between performance standards and organizational commitment.

Method

Sample and procedure

Data for this study were collected from employees working in Taiwanese firms manufacturing electrical and electronic products. I first contacted the HR managers at the related firms to ask them to participate in this study. Thirty-two HR managers stated that their companies were willing to participate in this work. Next, I mailed 623 questionnaires to the HR managers at these firms and asked them to arrange for some of their employees to complete the survey. Of the 623 questionnaires that were sent out, 395 complete and valid ones were returned, giving in an overall response rate of 71%. Most of the respondents worked as engineers (75%), and most had been with their current employer for more than one year (88%). Approximately 69% of the respondents were male. Eighty-seven per cent of the respondents were aged from 21 to 40 years. Seventy-seven per cent of them had an education level of college graduate or above.

Measures

The variables used in the current study are described below. Based on the related theories and previous research, a survey questionnaire was developed to collect data that could then be used to examine the validity of the model and hypotheses. The survey instrument included psychometric scales designed to measure administrative PA practices, organizational justice and organizational commitments. The survey questions were scored on a seven-point Likert-type scale, ranging from 1 = strongly disagree to 7 = strongly agree, to make survey administration easier and provide consistency in terms of response format. The surveys were administrated in Chinese, with survey items first translated into Chinese and then back translated to English to reduce the risk of meaning being lost. The measures used in this work have been translated into Chinese and used in many Chinese-language studies (e.g. Chou 2008) and thus should be as applicable to the mainly Taiwanese sample used in this study as they are to an English-speaking one.

Administrative PA practices

Salary adjustments. Five items ($\alpha = 0.92$) were developed to measure the PA practices related to salary adjustments based on a review of previous studies (Folger and Konovsky 1989; Anthony et al. 1993; Milkovich and Newman 1993; Bloom 1999; Milliman et al. 2002). Employees were asked to indicate the extent to which they agreed with each

statement, and an example item from this category is: 'I think my company is able to adopt objective job information to make salary decisions'.

Promotion decisions. Five items ($\alpha = 0.88$) were used to measure the PA practices related to promotion decisions based on previous research (Greenberg 1986; Milliman et al. 2002; Nurse 2005; Tekleab et al. 2005). An example item from this category is: 'I think every employee in my company is judged using the same promotion standards'.

Performance standards. Five items ($\alpha = 0.87$) were used to assess the PA practices related to performance standards based on prior research (Varma et al. 1996; Erdogan 2002; Holbrook 2002; DeNisi and Pritchard 2006; Cropanzano et al. 2007). An example item from this category is: 'I think every employee in my company is evaluated with the same performance standards' (Table 1).

Organizational justice

The measurements for organizational justice are divided into three constructs, distributive justice, procedural justice and interactional justice (Table 2), based on the studies conducted by Kang (2007), Moorman (1991) and Niehoff and Moorman (1993). Four items were used to measure distributive justice ($\alpha = 0.90$), with the respondents asked to indicate the extent to which they agreed with each statement. An example item for measuring distributive justice is: 'My work schedule is fair'. Five items ($\alpha = 0.90$) were

Table 1. Measurements of administrative PA practices.

<i>Administrative PA practices</i>	<i>Measuring items</i>
Salary adjustment	1. The pay I get in my company is related to my performance 2. I think my company offers me a better salary compared to what I would receive at other companies 3. I think my company is able to use objective job information when making salary decisions 4. I have an opportunity to express my opinions about salary decision to the management in my company 5. My company will consider employees' opinions when making salary decisions
Promotion decision	1. I think every employee in my company is judged using the same promotion standards 2. I think that employees who have excellent performance ought to be promoted in my company 3. I can question promotion decisions through a formal communication channel in my company 4. I think my company will follow the stated promotion standards to make promotion decisions 5. My company reveals all information related to job promotions to its employees
Performance standards	1. I think every employee in my company is evaluated with the same performance standards 2. I think the supervisors do use the performance standards made by the company to evaluate their subordinates 3. The standards used for performance appraisals are explained clearly to employees before being implemented in my company 4. My supervisor clearly explains the performance standards used to subordinates 5. I think that building a good relationship with my supervisor is helpful to obtaining a good result of my performance appraisal

Table 2. Measurements of organizational justice.

<i>Organizational justice</i>	<i>Measuring items</i>
Distributive justice	1. My work schedule is fair 2. I think that my level of pay is fair 3. Overall, the rewards I receive here are quite fair 4. I feel that my job responsibilities are fair
Procedural justice	1. Job decisions are made by the general manager in an unbiased manner 2. My general manger makes sure that all employee concerns are heard before job decisions are made 3. My general manager clarifies decisions and provides additional information when requested by employees 4. All job decisions are applied consistently across all affected employees 5. Employees are allowed to challenge or appeal job decisions made by the general manager
Interactional justice	1. When decisions are made about my job, the general manger treats me with kindness and consideration 2. When decisions are made about my job, the general manager treats me with respect and dignity 3. When decisions are made about my job, the general manager is sensitive to my personal needs 4. The general manager discusses the implications of any decisions related to my job with me 5. My general manager very clearly explains any decision made about my job

used to measure procedural justice, and an example items is: ‘Job decisions are made by the general manager in an unbiased manner’. Interactional justice was measured with five items ($\alpha = 0.91$) with an example item being: ‘When decisions are made about my job, the general manager treats me with kindness and consideration’.

Organizational commitment

A shortened version of the organizational commitment survey (Allen and Meyer 1990) was applied to measure the employees’ organizational commitment (Table 3), and the seven items with highest loadings were chosen for inclusion ($\alpha = 0.88$). A representative

Table 3. Measurements of organizational commitment.

	<i>Measuring items</i>
Organizational commitment	1. I feel ‘emotionally attached’ to this organization 2. This organization has a great deal of personal meaning for me 3. I feel a strong sense of belonging to my organization 4. It would be very hard for me to leave my organization right now, even if I wanted to 5. One of the few serious consequences of leaving this organization would be the scarcity of available alternatives 6. I believe that a person must always be loyal to his or her organization 7. One of the major reasons I continue to work for this organization is that I believe that loyalty is important and therefore feel a sense of moral obligation to remain

item is: ‘One of the major reasons I continue to work for this organization is that I believe that loyalty is important and therefore feel a sense of moral obligation to remain’.

Results

The means, standard deviations and correlations of the independent and dependent variables are presented in Table 4 with the Cronbach’s α for each scale shown in bold and on the diagonal. In general, the bivariate correlations provided confidence that the measures were functioning properly.

As suggested by Anderson and Gerbing (1988), this study followed the two-step structural equation modeling approach: a measurement model and a structural one. These multiple-indicator models were estimated by the maximum likelihood method using the LISREL 8.50 software package. The χ^2 statistic for the full measurement model was 1012.11 with 413 degrees of freedom (df), the goodness of fit index (GFI) was 0.86, the adjusted goodness of fit index (AGFI) was 0.83, the comparative fit index (CFI) was 0.99, the incremental fit index (IFI) was 0.99, the standardized root mean square residual (SRMR) was 0.039 and the root mean square error of approximation (RMSEA) was 0.061. All items significantly loaded onto their expected factors. Accordingly, the full measurement model provided an acceptable fit to the data and justified the use of the two-step approach.

The results of the structural model showed a reasonable fit to the data ($\chi^2 = 1049.69$, $df = 422$; GFI = 0.85; AGFI = 0.83; CFI = 0.99; IFI = 0.99; SRMR = 0.044; RMSEA = 0.061) and all the hypothesized paths in the proposed model were statistically significant ($p < 0.01$). Each standardized path coefficient for the structural model is presented in Figure 2.

I first tested the relationship between the administrative PA activities and organizational justice. Based on the results shown in Figure 2, salary adjustment decisions were significantly related to distributive justice but not with procedural justice, which supports H1.1 but rejects H1.2. Next, promotion decisions were significantly associated with distributive and procedural justice, supporting both H2.1 and H2.2. Third, an employee’s knowledge of performance standards was significantly correlated to procedural and interactive justice, supporting H3.1 and H3.2.

I also tested the relationships between organizational justice and organizational commitments. According to the results presented in Figure 2, organizational justice, including the constructs of distributive, procedural and interactional justice, was significantly associated with organizational commitment, supporting H4, H4.1, H4.2 and H4.3.

Table 4. Correlations between study variables.

Variables	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7
SA	3.66	1.49	0.92						
PRO	3.84	1.30	0.71**	0.88					
STD	3.86	1.25	0.67**	0.67**	0.87				
DJ	3.79	1.29	0.74**	0.68**	0.69**	0.90			
PJ	4.03	1.30	0.68**	0.68**	0.70**	0.66**	0.90		
IJ	4.16	1.33	0.65**	0.63**	0.70**	0.68**	0.69**	0.91	
OC	4.07	1.10	0.68**	0.70**	0.68**	0.70**	0.69**	0.71**	0.88

Note: SA, salary adjustments; PRO, promotion decisions; STD, performance standards; DJ, distributive justice; PJ, procedural justice; IJ, interactional justice; OC, organizational commitment. *N* = 395.

**Significant at the 0.01 level (two-tailed).

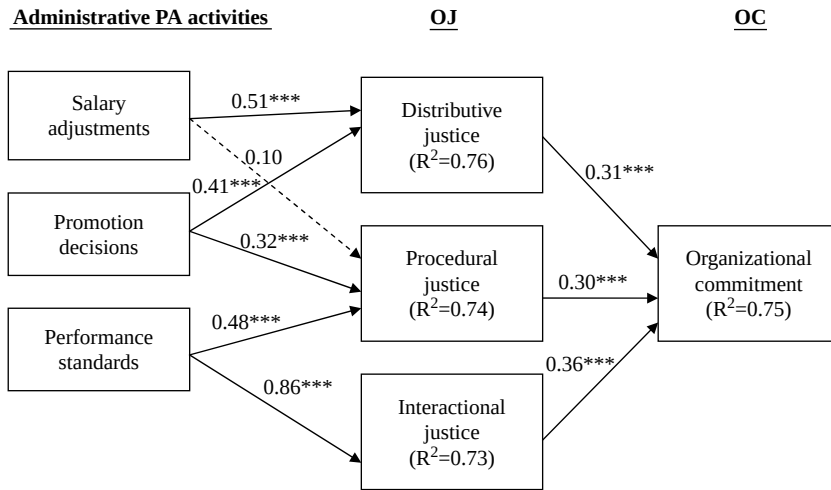


Figure 2. Standardized path coefficients for the structural model. Note: $N = 395$. *** $p < 0.001$.

Finally, I evaluated the hypotheses in which organizational justice was introduced to mediate the relationship between the administrative PA activities and organizational commitment using the testing suggested by Kenny, Kashy and Bolger (1998). In the first step, administrative PA activities must relate to organizational commitment, which is not shown in the research model. Second, administrative PA activities need to be correlated to organizational justice, which is supported by H1.1, H2.1, H2.2, H3.1 and H3.2, but not by H1.2. Third, organizational justice has to be associated with organizational commitment, which is supported by H4.1, H4.2 and H4.3. According to Kenny et al. (1998), the paths from administrative PA activities to organizational justice and from organizational justice to organizational commitment imply the path from administrative PA activities to organizational commitment. Since this research did not test whether there was any reduction in the direct effects of administrative PA activities on organizational commitment, the results are interpreted as indicating partial mediation.

In order to assess this partial mediation, Sobel's (1982) test was carried out to examine hypotheses 5.1, 5.2, 6.1, 6.2, 6.3 and 7.1. Hypotheses 5.1 and 5.2 are supported ($z = 4.19$, $z = 3.84$, $p < 0.001$), indicating that perceived distributive justice partially mediates the relationships between the decisions related to salary adjustments and promotions, and organizational commitment. Hypothesis 6.1 does not need to be examined because H1.2 is rejected. Hypotheses 6.2 and 6.3 are supported ($z = 2.69$, $z = 3.16$, $p < 0.01$) showing that perceived procedural justice has a partial mediating effect on the relationship between promotion decisions and performance standards, and organizational commitment. Hypothesis 7.1 is supported ($z = 5.66$, $p < 0.001$) indicating that the relationship between an employee's knowledge of performance standards and their organizational commitment is partially mediated by perceived interactional justice.

Discussion

This research found that perceived distributive justice can be predicted by salary adjustments and promotion decisions; perceived procedural justice can be predicted by

promotion decisions and performance standards; finally, perceived interactional justice can be predicted by performance standards. Furthermore, the results also showed that perceived organizational justice can predict the level of organizational commitment among employees. Most importantly, this study found that perceived organizational justice has a mediating effect on the relationship between administrative PA practices and organizational commitment, which has not been examined in previous studies. These findings add to our understanding of how PA-related HRM activities contribute to employee perceptions of organizational justice and organizational commitment in a number of ways, as explained in more detail below.

First, the finding that salary adjustment decisions have a positive association with perceived distributive justice is consistent with the predictions of Folger and Konovsky (1989). However, while Folger and Konovsky (1989) also indicated the salary adjustment decisions can significantly influence perceived procedural justice, the present research did not find this based on the data examined. One possible explanation for this is that Taiwanese employees often see salary adjustments as results-oriented (i.e. in terms of how much their organization pay them) rather than procedure-oriented (i.e. how payment decisions are arrived at) and this might alter the effects of that such decisions have on perceived procedural justice.

The findings of the present study also provide strong support for previous works which found that promotion decisions are closely associated with the employee perceptions of distributive and procedural justice (Mount 1983; Greenberg 1986; Colquitt et al. 2001; Nurse 2005; Tekleab et al. 2005; Jawahar 2007) and that performance standards also have a significant influence on the perceived procedural and interactional justice (Alexander and Wilkins 1982; Cardy and Dobbins 1986; Tsui and Barry 1986; Varma et al. 1996; Erdogan 2002). In addition, it is not surprising that the findings of this work are consistent with those of prior research which indicated that employee perceptions of organizational justice have a strong effect on the level of organizational commitment (Alexander and Ruderman 1987; Folger and Konovsky 1989; Konovsky and Cropanzano 1991; McFarlin and Sweeney 1992; Sweeney and McFarlin 1993; Lowe and Vondanovich 1995; Randall and Mueller 1995; Martin and Bennett 1996; Fields et al. 2000; Konovsky 2000; Masterson et al. 2000; Colquitt et al. 2001; Aryee et al. 2002; Kernan and Hanges 2002; Simons and Roberson 2003; Hui and Rupp 2005; Olkkonen and Lipponen 2006; Lavelle et al. 2007).

Finally, and most importantly, no research has yet investigated whether organizational justice acts as a possible mediator in the relationship between administrative PA and organizational commitment. The present results provide empirical support suggesting that employee perceptions of organizational justice are responsible for the impact of administrative PA on organizational commitment. In other words, the implementation of administrative PA affects organizational commitment through perceived organizational justice.

Implications

The findings of this work have a number of theoretical implications. First, while the effects of administrative PA practices (e.g. salary adjustments, promotion decisions and performance standards) on work attitudes and behaviors have attracted widespread research attention, few works have explored the relationships by which such practices influence employee attitudes and behaviors. The findings of this work suggest a mechanism for explaining the effects of administrative PA practices on employee perceptions of organizational justice and their level of organizational commitment.

Specifically, perceptions of fairness have a partial mediating role in the relationship between the implementation of various administrative PA practices and employee attitudes and behaviors.

Second, although researchers in organizational justice have differentiated between three distinct dimensions (i.e. distributive, procedural and interactional), much of the empirical work has examined only one or two types of it (Cohen-Charash and Spector 2001). This research included all three dimensions of organizational justice in one study and measured them separately, thus adding the literature with regard to the antecedents and consequences of organizational justice.

Moreover, previous research regarding to the relationships among PA, organizational justice and organizational commitment has focused on the relationship between either two constructs, such as the PA and organizational justice (e.g. Folger and Konovsky 1989), or between organizational justice and organizational commitment (e.g. Konovsky and Cropanzano 1991). This research investigated the three constructs in a holistic way, thus providing a more comprehensive understanding of the relationships among administrative PA, organizational justice and organizational commitment, and the mediating role that organizational justice plays in the relationship between administrative PA and organizational commitment.

Finally, the findings suggest guidelines for HR managers on how to promote employee perceptions of organizational justice and organizational commitment through the development of administrative PA programs. First, employees who recognize the implementation of salary adjustment decisions reported a higher level of distributive justice, but no effects were seen with regard to procedural justice. This suggests that when HR managers make salary-related decisions, they should be aware that the results may have a critical impact on employee perceptions of distributive justice and thus influence their organizational commitment. HR managers should therefore endeavor to make the process underlying such decisions more transparent to employees, and the focus should thus not only be on how much the organization pays, but also on the fairness of how such decisions are made.

Next, the results of this study also show that employees who recognize the implementation of promotion decisions also reported greater perceived distributive and procedural justice. This suggests that employees are more concerned with the processes underlying promotion decisions than they are with those related to salary adjustments. Since promotions are generally associated with pay rises, this implies that HR managers should consider internal candidates over external ones when there is a job opening in the organization, especially for management positions, as doing this could enhance employee perceptions of organizational justice, and in turn increase their organizational commitment.

Finally, the results of this study show that the greater the knowledge with regard to performance standards, the greater the perceptions of procedural and interactional justice among employees. They also show that employee's knowledge of performance standards has a stronger effect on employee perception of interactional justice than procedural justice. Performance standards can be seen as a type of psychological contract, and from this perspective employees agree to meet these standards in order to be paid, while the organization is willing to pay its employees in order to have this goal. The more specific the contract, the more positive employee perceptions with regard to organizational justice. This implies that HR managers should work to increase effective communication and employee involvement within the organization (Zhang and Agarwal 2009), as this can make the expected standards more transparent, thus increasing the perception of organizational justice among employees.

Limitations and future research

Like other research, this study is not without limitations, which present some avenues for future work. First, the data used in this study were collected from a single source and thus there is a possibility that the present findings may be partly affected by common method bias. Although the results of confirmatory factor analysis showed that no general factor accounted for most of the covariance in the dependent and independent variables, thus indicating that common method bias had minimal effects on the results (Podsakoff and Organ 1986), collecting data from one source is still a limitation. Therefore, it would be of interest if future research could reduce this risk by asking questions at different points in time or designing separate questionnaires to collect data from multiple sources.

Second, generalization of the findings is another limitation of this work. The respondents in this study were all employees at Taiwanese manufacturing companies producing electrical and electronic goods and thus the results should be extended to other groups with caution. Accordingly, this study needs to be replicated in other kinds of companies, such as those in service industries. Moreover, it would be of interest for future research to examine the relationships among PA practices, organizational justice and organizational commitment in different countries.

Finally, this study only examined three types of PA practices for administrative purposes, but other practices, such as documentation or PA interviews (Pettijohn et al. 2001; Milliman et al. 2002; Wright 2004), would also affect employee perceptions of organizational justice. It would thus be of interest if future research could examine the effects of developmental PA practices, including training and development activities, as well as performance feedback, on organizational justice and organizational commitment.

Conclusions

The purpose of this study was to test the relationships among administrative PA activities, perceived organizational justice and organizational commitment. The results showed that employee perceptions of organizational justice could be predicted by the focal PA practices. The greater the implementation of these practices, the greater the perception of organizational justice, which in turn can increase the level of organizational commitment. Therefore, organizations should aim to carry out clearer PA activities in order to enhance the perception of organizational justice and thus increase organizational commitment among employees.

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